Does Cropping the Dog Help?
A Study of Background Removal in Fine-Grained Breed Classification

Wuqian Ma 02-05-2026

Abstract. This study asks if automatically cropping the dog with the open-vocabulary detector Grounding-DINO [4] can improve fine-grained breed classification on the Stanford Dogs / Kaggle Dog Breed dataset [2, 1] (10 222 training images, 120 breeds). I compare four runs that vary one design lever at a time, from a frozen ResNet-50 baseline to a fine-tuned ConvNeXt-Base [3] at 384 px operated on either full or cropped images. Cropping changes the test log-loss from 0.287 to 0.361 (a relative deterioration of 26%) and reduces top-1 accuracy by 3.2 percentage points; top-5 is approximately preserved. A per-class breakdown shows the loss is heterogeneous: only 17 of 120 breeds gain accuracy under cropping while 50 lose accuracy and the remaining 53 are unchanged, with the largest losses falling on visually near-identical breed pairs (*Staffordshire Bullterrier* vs. *American Staffordshire Terrier*; *Siberian Husky* vs. *Malamute*; *Greater Swiss Mountain Dog* vs. *Appenzeller*). The background therefore appears to carry context that the network uses to disambiguate look-alike pairs, and foreground segmentation is not the appropriate lever to reduce the residual errors on this dataset.

1 Problem formulation

The Stanford Dogs / Kaggle Dog Breed dataset contains 10 222 training images across 120 breeds (66–126 per class, 1.91× imbalance). Two sources of difficulty motivate the study: (i) the images contain substantial irrelevant background (humans, accessories, clutter), and (ii) several breed pairs are visually near-identical (e.g. *Siberian Husky* / *Alaskan Malamute*, *Great Pyrenees* / *Kuvasz*, *American* / *Staffordshire Bullterrier*). The question I address is whether automatic cropping improves classification, or whether the look-alike problem is intrinsic to the foreground and dominates any background gain.

2 Model design choices

I compare four runs that differ in exactly one lever each, so that every difference is attributable to a single design decision (Tab. 1). The ladder isolates pretraining alone (E0), end-to-end fine-tuning (E0→E1), backbone capacity and input resolution (E1→E2), and the cropping intervention itself (E2→E3). Cropping uses Grounding-DINO with the breed-agnostic prompt `"a dog ."`, box threshold 0.30, and a 10% margin around the highest-confidence box; full-image fallback fires on 5/9152 processed images (0.05%). A breed-specific prompt would leak labels into the cropping pipeline and is deliberately avoided.

Table 1: Experimental ladder. Each row adds a single lever relative to the previous; only E2 vs. E3 differs in the input pixels.

Run	Backbone	Resolution	Image	Trainable parameters	Question
E0	ResNet-50 (frozen) + linear head	224	full	head only	ImageNet features alone
E1	ResNet-50 fine-tuned	224	full	all	Gain from end-to-end fine-tuning
E2	ConvNeXt-Base `fb_in22k_ft_in1k_384`	384	full	all	Stronger backbone, higher resolution
E3	ConvNeXt-Base `fb_in22k_ft_in1k_384`	384	cropped	all	Principal comparison: crop vs. full

3 Training strategy

A single 60/20/20 stratified split (seed 42) yields 6133/2044/2045 train/val/test images. Augmentation is fixed across all runs to keep the ladder comparable: RandomResizedCrop, horizontal flip, mild colour jitter, RandAugment (m=7), MixUp (α=0.2) with CutMix (α=1.0), RandomErasing (p=0.25), and label smoothing 0.1. Optimisation is AdamW with cosine decay, gradient clipping at 1.0, and AMP fp16. At inference, I reload the best-validation checkpoint, collect probabilities with horizontal-flip test-time augmentation on E2/E3, fit a single temperature $T^\star$ [5] on validation log-loss, and apply it to the test logits. Per-run wall-clock on a single A100-80GB: 4–7 min for E0/E1 and approximately 45 min for E2/E3.

4 Hyperparameter tuning

A five-run sweep on E1 (5 epochs, val log-loss; head LR fixed at 10× base LR) covered the LR / weight-decay grid, with the following best-validation log-losses: (base, head, wd) = $(10^{-4}, 10^{-3}, 0.05) \to 0.61$; $(10^{-4}, 10^{-3}, 0.01) \to 0.69$; $(3{\cdot}10^{-5}, 3{\cdot}10^{-4}, 0.01) \to 3.30$; $(3{\cdot}10^{-5}, 3{\cdot}10^{-4}, 0.05) \to 3.38$; $(10^{-5}, 10^{-4}, 0.05) \to 4.46$. The first configuration is locked in for E1 and rescaled by 1/5 for E2 and E3 (a common fine-tuning practice when moving to a larger backbone at higher resolution). The fitted temperatures are $T^\star_{E1}$=0.62, $T^\star_{E2}$=0.72 and $T^\star_{E3}$=0.72; for E0 the L-BFGS solution is negative ($T^\star = -0.70$), which inverts the logit ordering — a scalar temperature is not expressive enough to calibrate an uncalibrated frozen-features softmax — so the raw E0 row is reported in Tab. 2.

5 Performance analysis

Principal comparisons. End-to-end fine-tuning (E0→E1) reduces test log-loss by a factor of 2.9 at essentially the same top-1 accuracy. Replacing the backbone (E1→E2) yields a small further reduction (test log-loss 0.290 → 0.287, top-1 91.25% → 91.88%). The cropping intervention (E2→E3) deteriorates test log-loss by $\Delta = +0.074$

Table 2: Test-set metrics. E1–E3 are reported after temperature scaling; E0 is reported on raw probabilities (see §4). Lower is better for log-loss and ECE; higher is better for the accuracies and macro-F1.

Run	Log-loss	Top-1	Top-5	Macro-F1	ECE
E0 (RN50 frozen)	0.836	0.9227	0.9936	0.9173	0.408
E1 (RN50 fine-tuned)	0.290	0.9125	0.9892	0.9070	0.012
E2 (ConvNeXt-B, full)	0.287	0.9188	0.9936	0.9137	0.018
E3 (ConvNeXt-B, cropped)	0.361	0.8870	0.9917	0.8816	0.024

(relative +26%) and top-1 accuracy by 3.2 percentage points; top-5 is approximately preserved (0.9936 → 0.9917), which suggests that cropping shifts probability mass between similar candidates rather than ejecting the truth from the short list. The hypothesis that the background was a confounder for breed identity is therefore not supported on this dataset.

Per-class breakdown (Fig. 1). The aggregate result hides a heterogeneous pattern. The gains cluster on centrally framed dogs, where cropping appears to yield a sharper view of texture and silhouette; the losses cluster on look-alike pairs flagged a priori in the EDA, where the full-image model appears to have used contextual cues (typical surroundings, scale relative to in-frame objects, photographic conventions) that disappear inside a tight bounding box.

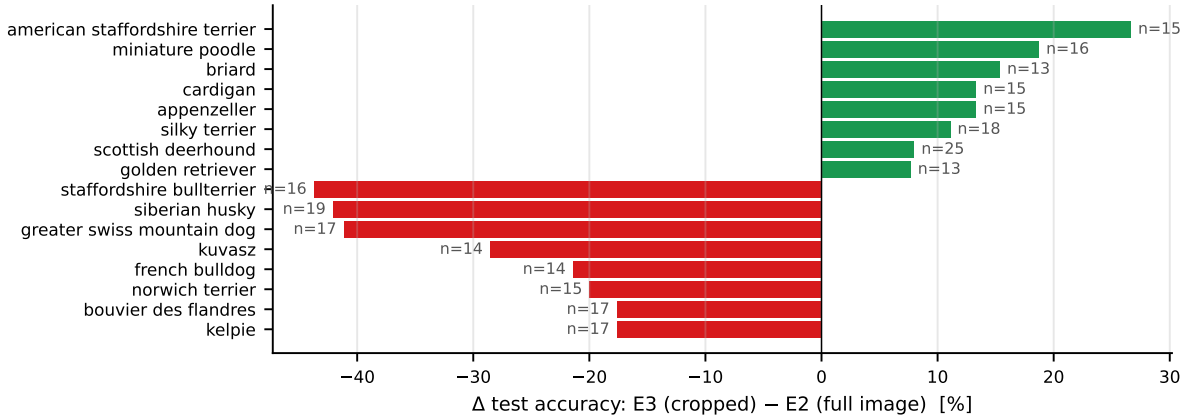


Figure 1: Per-class effect of cropping on test accuracy (E3 minus E2). Each horizontal bar is one breed; n is the number of test images for that breed (mean ≈17, range 13–25). *Upper eight bars (green):* the breeds that gained the most accuracy from cropping, all of them centrally framed dogs where cropping yields a sharper crop of texture and silhouette. *Lower eight bars (red):* the breeds that lost the most, all of them members of visually near-identical breed pairs whose disambiguation appears to have relied on the background. Across all 120 breeds, 17 gained, 50 lost, and 53 were unchanged. A single-image flip corresponds to ∼5–8 percentage points; the four largest losses correspond to 7 or more misclassifications out of ≤ 19 test images for the breed and therefore exceed plausible sampling noise.

Confusion structure. The persistent confusion of *eskimo_dog* as *siberian_husky* (12 misclassifications out of 13 test images of *eskimo_dog* in E2; 10 out of 13 in E3) reflects label inconsistency in ImageNet itself. Cropping then introduces three additional high-frequency confusions that were absent or only mild in E2: 8 of the 16 test images of *Staffordshire Bullterrier* are misclassified as *American Staffordshire Terrier*; 6 of 17 *Greater Swiss Mountain Dog* images are misclassified as *Appenzeller*; and 6 of 19 *Siberian Husky* images as *Malamute*. These three pairs — all of them flagged a priori in the EDA as visually similar — account for the bulk of the test-set deterioration of E3 relative to E2.

6 Limitations

The dataset is a curated subset of ImageNet-1k [2], so absolute scores reflect partial memorisation by the pre-trained backbone; the inter-run deltas share the same leakage and remain interpretable, while absolute claims about generalisation are not warranted. The single 60/20/20 split gives approximately 17 test images per class, so single-image flips correspond to ∼6 percentage points and individual-breed comparisons are noisy (the aggregate deltas are not). Crop quality was inspected on a 50-image visual sample only, and the comparison is restricted to one backbone family (CNNs). The most informative future intervention therefore appears to be model-side — a metric-learning auxiliary loss on hard pairs, or a high-resolution head on faces and ears — rather than further input pre-processing.

References

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